Arjit Yadav

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PROFILE

Backend and AI product engineer with an M.Sc. in Computer Science (Universität Stuttgart, 1,9) who ships LLM features end to end for regulatory and compliance work. I built and run a live EU AI Act copilot on my own: it turns a company's description of an AI system into its obligations, deadlines and gaps, using a LangGraph agent, structured outputs, hybrid retrieval over the full regulation on Postgres, citations checked in code, and a 20-scenario eval suite that decides what ships. At Valeo I built the tool engineers use to ingest specifications and reports as Excel, CSV or PDF, validate them, and flag inconsistencies with Vertex AI. Before that I maintained the REST services and MongoDB collections behind a product with millions of listings at a small SaaS company, with CI/CD on GitHub Actions and Docker. I talk to users first and measure whether what I shipped worked.

EDUCATION

M.Sc. Computer Science 10/2023 - 05/2026
Universität Stuttgart, Germany Grade: 1,9

- Major: Autonomous Systems
- Thesis: Energy-Efficient Semantic Segmentation using Spiking Neural Networks (SNNs)

B.Tech. Information Technology 07/2017 - 07/2021
University of Delhi, India Grade: 2,1

PROJECTS

EU AI Act Compliance Copilot
Multi-agent RAG system for regulatory compliance | Live demo | GitHub

- Built a system that classifies AI systems under the EU AI Act and maps their obligations, deadlines and compliance gaps: a LangGraph pipeline with a clarifying-question step, parallel LLM and ML classification, and a writer-verifier loop capped at three attempts.
- Hybrid retrieval over the full Act (292 structure-aware chunks) with pgvector, Postgres full-text search and Reciprocal Rank Fusion reached recall@5 of 0.90, against 0.42 for keyword search alone. Citations, legal rules and deadlines are checked in code rather than trusted from the model, with prompt-injection and PII guardrails in front.
- Used a 20-scenario eval suite and error analysis to drive fixes, not guesswork: one run showed the verifier rejecting correct reports, and moving that step into code cut assessment time by 57%.
- Trained an Annex III classifier as an independent second opinion (scikit-learn, MLflow registry with a promotion gate, drift monitoring). Reviewing 270 synthetic training rows by hand - 43 relabelled, 72 removed as wrong or leaking - took macro F1 from 0.22 to 0.73.
- Deployed for $0 on Oracle Cloud with Docker Compose, Caddy (HTTPS) and a Streamlit UI, using gpt-oss-120b on Groq. GitHub Actions runs integration tests against real Postgres and Redis, and the server deploys green commits itself, rolling back automatically when a release is unhealthy.

PROFESSIONAL EXPERIENCE

Software Engineer - Full-Time 06/2026 - Present
SWJ Germany GmbH, Bietigheim, Germany

- Working on an LLM-driven regression test selection system for automotive software: when the code changes, it works out which tests actually need to run, and in what order.
- Built the Python ETL pipeline that pulls requirements, test cases and task descriptions from Codebeamer (ALM) on every run and filters them down to what the project's rules consider relevant.
- Wrote the pipeline that pulls the GitHub repository with Git and extracts the code diff between software versions.
- Built a code impact module on Google Gemini that reads each task description alongside its diff and identifies which parts of the software the change touches.

Working Student - Part-Time 11/2025 - 04/2026
Valeo Schalter und Sensoren GmbH, Bietigheim, Germany

- Built the tool the team uses to validate quality reports: a Streamlit interface where engineers upload the relevant documents - test and requirement specifications, traceability matrices, test reports - as Excel, CSV or PDF.
- Wrote the Python backend behind it. It checks each upload is the document it claims to be and complete enough to work with, then cleans and filters the contents against the team's rules before any comparison happens.
- Google Cloud Vertex AI does the final pass, checking the test report against what the specifications require and flagging where the two disagree. That cross-check used to be done by hand.

Working Student - AI & Computer Vision - Part-Time 07/2024 - 04/2025
Vreed Immobilien Digital, Stuttgart, Germany

- Fine-tuned VGG16 classifiers in PyTorch (transfer learning) to sort photos of WBS70 Plattenbau housing, reaching 78% accuracy on imagery that was never shot to any standard.
- Trained Detectron2 (R50-FPN) models to locate solar panels and window frames in facade images, feeding automated energy efficiency audits.
- Built a balcony detector from scratch, AP 54, so renovation planning data could be collected city-wide instead of surveyed by hand.

Software Engineer - Project-Based 04/2023 - 09/2023
JindalX, New Delhi, India

- Automated roughly 15 billing and claim-filing processes in healthcare revenue cycle management using Python, Selenium and PyAutoGUI, all of them previously done manually.
- Added retry logic and logging so failures surfaced instead of quietly corrupting patient billing records; data handling followed HIPAA requirements.
- Sat with the operations team to write down the business rules they had been following by habit, then turned those into the specs the automation ran on.

Software Engineer - Full-Time/Freelance 03/2022 - 09/2023
SaaS Platformen ApS, Remote

- Wrote the distributed scrapers, Python on AWS Lambda, that pulled car listings from 32 external sources and normalised them into one schema.
- Maintained the REST services and MongoDB collections behind the product: millions of listings, with prices and availability changing constantly.
- Built the alerting side on AWS SNS/SQS so users were notified on price drops and new matches against their saved filters.
- Set up CI/CD with GitHub Actions and Docker, along with health checks, which took manual steps out of every release.

Data Scientist - Internship/Full-Time 04/2021 - 10/2021
Pivony, Remote

- Built semantic document search on MPNet (BERT) embeddings with cosine similarity, which caught paraphrased queries the existing keyword search missed.
- Put together a Turkish NLP pipeline that extracted addresses from free text and resolved them to GPS

coordinates using a Turkish cities database and mapping APIs.
- Wrote the topic labelling step on top of LDA, using n-grams to turn term distributions into short multi-word labels a person could actually read.
- Owned the REST APIs that served all of this, including maintenance after handover.

Data Analyst Intern & Mentor 06/2019 - 08/2019
Aam Aadmi Party - Data Analysis Division, New Delhi, India

- Wrote the Python scraping and OCR pipeline that turned thousands of scanned voter list PDFs into structured records.
- Built a Python library the division used for constituency-level breakdowns by gender, income bracket and caste.
- Co-led mentoring for a cohort of around 100 interns: technical onboarding, code review, and keeping the data consistent across teams.

TECHNICAL SKILLS

Languages: Python (Primary), SQL, JavaScript, Java, C/C++, Bash, R
Applied AI / LLM: RAG (Hybrid Retrieval, Reciprocal Rank Fusion, Grounded Citations), Tool-Using Agents (LangGraph, LangChain), Structured-Output Pipelines (Pydantic), Evaluation Workflows, Guardrails; Google Gemini, Vertex AI, OpenAI-Compatible APIs, Anthropic Claude
Backend: FastAPI, Pydantic, REST, Django, Flask, Redis/RQ (Background Queues), API and Data-Model Design
Data: PostgreSQL (pgvector, Full-Text Search), MongoDB, Redis, DynamoDB, ETL Pipelines, Document Parsing (PDF/Excel/CSV)
Regulatory & Reporting: EU AI Act Obligations and Deadlines, Requirements Traceability, Quality-Report Validation, HIPAA-Compliant Data Handling
Cloud & Quality: Docker, GitHub Actions (CI/CD), AWS (Lambda, S3, SNS/SQS), GCP, Oracle Cloud, PyTest, Integration Tests, Prometheus, Grafana, MLflow
ML: PyTorch, scikit-learn, TensorFlow, OpenCV, Detectron2

LANGUAGES

English: Fluent | **German:** A2